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Predictors of community empowerment against dengue in Córdoba, Colombia

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Abstract

Introduction Given the phenomena and problems caused by neglected tropical diseases such as dengue, questions arise about the effectiveness of policies that guide control and prevention strategies in the territories at greatest risk, demanding greater research efforts. This study evaluates participation and governance as probable factors of community empowerment to address dengue in the department of Córdoba, Colombia.

Method A cross-sectional study was conducted between 2022 and 2024. A multistage cluster probability sampling method was used to select four municipalities, rural and urban sectors, households, and 352 individuals out of a total of 4,169 people infected with dengue in 2019. Data were collected using a structured questionnaire that included a scoring system to measure empowerment at high, low, and medium levels. An ordinal regression model was also defined to identify predictors of empowerment, including variables such as sociodemographic characteristics, knowledge about dengue, sanitation skills and conditions, participation, and governance, assigning a score to each response using the Likert scale technique.

Results The study found that as people participate more actively in processes related to their well-being or community, they have a greater perception of the importance of participation, communication, and management of dengue prevention and control programs, as well as greater access to information, better prevention practices, and better sanitation conditions in the home. The probability of achieving a higher level of empowerment doubles; (OR = 1.97; p-value < 0.0001), compared to those with lower levels of the same characteristics or factors.

Conclusions There is a need to strengthen inclusive, intersectoral community strategies with a territorial and rights-based approach to comprehensively address dengue in high-risk areas. This is in order to promote empowerment based on dialogical, participatory, and community management processes to improve disease prevention and control practices.

Keywords Empowerment, Participation, Dengue, Public health policy, Governance

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Introduction

Dengue is a neglected tropical disease (NTD) that poses challenges to public health policies around the world every year due to the complexity of its approach, its multifactorial nature, and the high human, social, and economic burden it represents for vulnerable communities [1, 2]. In this regard, the World Health Organization (WHO) highlights that one of the best bets for global public health is NTD prevention and control interventions, as these are an indicator of interest in detecting disparities in progress toward universal health coverage and access to equitable, high-quality health services [1].

The lessons learned in the fight against dengue fever show how the conceptual and methodological platforms that guide control and prevention programs are being strengthened with the incorporation of innovative technologies, genetic analysis, and the development of antiviral drugs and vaccines, which show promising advances in reducing viral load and immune function [3–9]. Despite this, knowledge about the sociocultural, organizational, and participatory characteristics of the population as potential factors for empowerment is still insufficient, requiring diverse epistemological and methodological perspectives for a comprehensive approach to dengue from the territories [10].

Hence, there is a need for greater research efforts with an emphasis on global governance and public policy management [11, 12]; considering empowerment, within this context, a key factor in the processes of strengthening health and social protection, which involves providing communities with tools related to education, information, and resources so that they can actively participate in their health care and influence health policies to autonomously manage their social and economic well-being [13].

The behavior of dengue in Colombia keeps health authorities on alert due to the high incidence rates in the last decade, with notable records in 2019 (254.2 per 100,000 inhabitants), 2023 (253 per 100,000 inhabitants), and 2024 (613.3 per 100,000) [14]. At the same time, some departments throughout the country warrant special analysis due to the confluence of risk factors, such as Córdoba, located on the Colombian Caribbean coast, which has had very high epidemiological records in recent years [15].

The fight against dengue in Córdoba combines a multiplicity of risk factors linked to geographical, climatic, sociocultural, and political characteristics that favor the permanent circulation of dengue virus serotypes in the territory [16]. Among other factors, low levels of community management, lack of motivation, inadequate dengue control habits, and low community participation continue to be causes for concern, posing challenges for the

evaluation of educational interventions [17] or the management of empowerment strategies that.

contribute to improving disease control and prevention processes [18, 19].

Based on the above, this research was conducted to evaluate participation and governance as probable factors of empowerment to address dengue in the department of Córdoba, Colombia, under the premise of conceiving empowerment as a process through which communities acquire a set of skills, develop the confidence to make better decisions about their own environment, and adopt participatory behaviors, contributing their knowledge and experience and learning in the context in which they interact [18].

Materials and Methods

Study design

This was a descriptive study with a quantitative approach, using an analytical design based on an ordinal regression model aimed at identifying predictors of community empowerment.

Research context

The study was conducted in the department of Córdoba, located in the Caribbean region of Colombia, with a population of approximately 1,898,911 inhabitants distributed across 30 municipalities [15]. Four of these municipalities were selected based on epidemiological and sociodemographic criteria: Montería (highest number of dengue cases reported in the National Public Health Surveillance System (Sivigila)), Pueblo Nuevo (highest incidence rate per 1,000 inhabitants), San Andrés de Sotavento (predominantly indigenous population), and San Bernardo del Viento (predominantly Afro-descendant population).

Population and sample

The target population consisted of 4,169 people diagnosed with dengue during 2019 according to Sivigila records [20].

2023 ($P=0.003872$) of 387.2 cases per 100,000 inhabitants, with a confidence level of 99% and a margin of error of 1%. Additionally, a design effect ($deff=1.05$) corresponding to a 5% increase over the base sample was taken into account to compensate for possible errors associated with the sampling methodology for a final minimum sample of 254 individuals. Finally, 352 people were surveyed.

When analyzed by gender, the population of the municipalities included shows a distribution close to equilibrium, with 49.0% men and 51.0% women. However, according to the cases of dengue reported in Sivigila in 2019, the proportion was 46.5% and 53.5% for men and women, respectively. The sample obtained was made

up of 68.9% women and 31.1% men, which reflects the predominant role of women in caregiving, cleaning, and household management activities, who were the most accessible during the data collection process.

With regard to age, the target population is distributed as follows: 20.9% young,

people, 61.3% adults, and 17.9% older adults, while the sample reached 18.5% young people,

69.9% adults, and 11.6% older adults.

A statistical power analysis was performed using a t-test for independent samples, considering a moderate effect size ($d=0.5$) and a significance level of 1% ($\alpha=0.01$), which yielded a statistical power of 0.99997, indicating a high capacity of the study to detect significant differences.

The sample was determined by the multistage cluster probability sampling method. In the first stage, the municipalities were selected using the criteria noted above; in the second stage, within each municipality, an urban and rural zoning process was carried out to delimit and select the communes, neighborhoods, townships, and villages, using information from Sivigila.

A third stage involved the identification and random selection of dwellings registered.

Participants were selected using the following inclusion criteria: i) being over 18 years of age, ii) residing in one of the four selected municipalities, iii) at least one person in the household having suffered from dengue in 2019, and iv) having lived in the place of residence for at least two years. The exclusion criteria were: i) being a minor, ii) living temporarily in the municipalities, and iii) having physical, mental, or emotional limitations that prevented them from responding to the survey.

Data collection

Data were collected using a questionnaire consisting of 37 questions, divided into four dimensions: sociodemographic characteristics, knowledge about the disease, community participation, communication, education, and governance based on the level of empowerment.

The instrument underwent content validity testing by experts in the fields of public health, epidemiology, community nursing, and social psychology. Similarly, a pilot test was conducted with 30 subjects with characteristics similar to those of the sample in order to identify gaps or strengths in understanding the questions.

The main changes made after the pilot focused on the participation item, specifically the questions: Which of the dengue prevention and control programs offered by health and government entities have you participated in? and What actions do you take with your community to prevent and control dengue in your neighborhood? After including the indicator of mosquito breeding site eradication and requesting fumigation from the Ministry

of Health or another entity, respectively, because the respondents considered it relevant to find these options as they are common in the territory.

Additionally, the instrument was tested for validity and reliability. A Cronbach's alpha coefficient of 0.879 was obtained, indicating adequate internal consistency. Exploratory factor analysis yielded a KMO sample adequacy index of 0.597, and Bartlett's sphericity test was significant at $p<0.005$, indicating appropriate conditions for applying this analysis.

Participants were initially contacted by telephone to coordinate their availability in terms of time and space. During the face-to-face survey process, each person's willingness to participate was confirmed, the informed consent form was read, and, after acceptance, the information was collected by asking the questions in the questionnaire.

Study variables

The dependent variable was the "level of empowerment," operationalized as the sum of the scores obtained in the following dimensions:

Knowledge of the disease (concepts, causes, symptoms, and treatment, maximum 8 points) + (sanitation practices = infrastructure and sanitation services in the home, 11 points), **Community participation** (9 points), **and Perception** (communication, education, and governance (13 points). The independent variables included **sociodemographic characteristics** (gender, marital status, age, socioeconomic status, educational level, family structure, social stratum, ethnicity, religion, and income), as well as the dimensions of **perception of governance and participation**.

Data management and analysis

The chi-square test was used to explore the associations between the level of empowerment and the sociodemographic variables, with a significance threshold of 0.05. The normality of the age variable was assessed using the Shapiro–Wilk test, obtaining a result of $W=0.9695$ and a $p\text{-value}=9.697\text{e-}07$, which indicates that age meets the assumption of normality.

The Shapiro–Wilk test and the Lilliefors (Kolmogorov–Smirnov) test are noteworthy as statistically robust tools for evaluating the normal distribution of data. With regard to age, both tests presented very small $p\text{-values}$ ($9.70\text{e-}07$ and $5.78\text{e-}05$), suggesting that the null hypothesis of normality should be rejected. Given the above, it was considered appropriate to use non-parametric tests, such as the Kruskal–Wallis test, to compare numerical variables between different levels of categorical variables.

Empowerment was measured using a Likert scale structured in four dimensions, with each

valid response assigned a score based on the degree of complexity:

for incorrect responses or a null level (0), basic (1), medium (2), and high (3). The total score achieved was 41 points, distributed as follows: i) knowledge about dengue (8 points), ii) sanitation practices (11 points), iii) participation (9 points), and iv) perception of communication and governance (13 points).

Based on this score, three levels of empowerment were categorized: low, medium, and high, taking as a reference the proportion of the maximum possible score ($\leq 25\%$, $25-75\%$, and $\geq 75\%$). Although this scoring system does not follow a conventional standard, it is supported by the methodological experience of other studies [21–23].

Based on the above, an ordinal regression model was constructed, in which the level of empowerment (low, medium, high) was the dependent variable. In this model, the scores obtained in the participation and perception dimensions were statistically significant for the adjusted model. The assumption of odds proportionality was then verified using Brants test, which assesses whether the model coefficients are constant across the different levels of the dependent variable, and was found to be significant and valid in this case.

Regarding the consistency of the predictors, it was found that, in the three classification schemes (original, low, and high), both participation and perception are highly significant with p-values below 0.001.

The results of the sensitivity analysis show that the models conclusions are robust: regardless of the cut-off points chosen to classify empowerment, the variables Participation and Perception remain significant predictors with the same direction of effect.

In the case of sanitation, in the initial analysis, this variable showed an estimated coefficient of 0.0853 with a standard error of 0.0676, resulting in a t-value of 1.26.

Then, when attempting to include the knowledge variable in the ordinal regression model using the polar function, problems of collinearity and category separation were identified, which prevented the estimation of reliable parameters. The model yielded adjustment errors related to the absence of initial values, indicating that the data structure made its incorporation into the multivariate analysis unfeasible.

Data processing and analysis were performed using Excel (Microsoft 365), R Studio.

(version 4.4.6), and SPSS (Statistical Package for the Social Sciences, version 25).

Results

Of the 352 participants, 68.9% were women, ranging in age from 19 to 87 years (), with an average age of 41.6 years and a standard deviation of 14.2 years.

In order to identify which dimensions provide the most information for understanding the structure of the relationships between the variables analyzed, a Multiple Correspondence Analysis (MCA) was performed, the results of which are shown in Table 1:

Table 1 shows the proportion of variance explained by each dimension of the correspondence analysis in four columns. The first column indicates the components or axes derived from the MCA, the second reports the eigenvalues, which represent the size of the variance explained by each of the new dimensions or axes. The third column shows the percentage of total variance explained by each dimension or axis, and the fourth reflects the percentage of cumulative variance.

Regarding the variables included in this analysis, in addition to the sociodemographic characteristics, it was found that the first dimension explains 16.31% of the total variability; likewise, the first fifteen dimensions together explain 93.03% of this variability, while the first ten dimensions accumulate 75.26%.

These results suggest that most of the individuals included in the sample are grouped consistently with the response options of the instrument applied. However, some categories are underrepresented in the model, such as the "High" level of participation, the "Low" level of learning in the municipalities of "San Bernardo del Viento" and "San Andrés de Sotavento," and the "Rural" area of residence. Their representativeness is low, that is, their classification percentage was low. This indicates that these groups exhibit heterogeneous behavior with respect to the rest of the sample.

Table 1 Proportion of variance retained by each dimension

Dimension	Value	Percentage of variance	Percentage accumulated of variance
1	0.3425	16.3113	16.3113
2	0.2658	12.6564	28.9680
3	0.1718	8.1831	37.1512
4	0.1658	7.8967	45.0478
5	0.1198	5.7043	50.7522
6	0.1141	5.4323	56.1844
7	0.1083	5.1551	61.3395
8	0.1051	5.0026	66.3422
9	0.0940	4.4782	70.8204
10	0.0933	4.4410	75.2614
11	0.0905	4.3091	79.5705
12	0.0801	3.8132	83.3836
13	0.0766	3.6487	87.0323
14	0.0702	3.3413	90.3736
15	0.0558	2.6556	93.0292

Source: Own construction

Correlation levels of the variables with the dimensions

Table 2 shows the correlation levels of the variables with the dimensions obtained in the multiple correspondence analysis, allowing us to identify which variables are most strongly associated with each axis and, therefore, which contribute most to the construction of the factorial space.

Table 2 identifies the most relevant variables for the first dimension as: municipality of residence, ethnicity, sanitation, and level of education, with coefficients of determination (r^2) of 0.8753, 0.8391, 0.7109, and 0.3716, respectively. For dimension two, the most important variables were the municipality where the study was conducted, ethnicity, sanitation, and perception, with r^2 of 0.8381, 0.7585, 0.4692, and 0.1911.

Similarly, analysis of the categories of these variables showed that the 15 main categories for dimensions one and two are: Montería (as a municipality), ethnicity (none), high level of sanitation, origin (urban and rural), high level of learning and low level of sanitation, and indigenous ethnicity. In terms of dimension two, does not know/does not respond in ethnicity, the municipality of San Andrés de Sotavento, indigenous ethnicity, medium and low sanitation, the municipality of San Bernardo del Viento, high levels of learning and perception, the municipality of Pueblo Nuevo, medium learning and participation, and low participation and empowerment.

Quality of representation of categories

To determine the contribution of each category to the respective selected dimension, the cosine square measure

Table 2 Correlation levels of variables with dimensions

Variable	Dimension 1		Dimension 2	
	R ²	P-value	R ²	P-value
Municipality	0.8753	6.79E-157	0.8381	3.64E-137
Ethnicity	0.8391	3.04E-136	0.7585	1.19E-105
Sanitation	0.7109	9.08E-95	0.4692	9.97E-49
Learning	0.3716	6.28E-36	0.1107	1.29E-09
Origin	0.2644	3.80E-25	0.0673	8.01E-07
Empowerment	0.1412	2.89E-12	0.1108	1.25E-09
Gender	0.0860	2.07E-08		
Skills	0.0802	4.64E-07	0.0311	4.05E-03
Perception	0.0389	9.87E-04	0.1911	8.48E-17
Participation	0.0179	4.28E-02	0.0807	4.18E-07

Source: Own construction

was applied, whereby if a category or variable has a high $\cos_2$ on the first axis, it means that that category is strongly related to the first dimension extracted and, therefore, its position in the graph is reliable in relation to that dimension. In this way, it was identified that the first two dimensions explain 28.97%. (fig. 1).

Figure 1 shows that the categories (response options) that are most important in the first dimension are ethnic group, municipality, and sanitation level. The upper right part of dimension 1 (quadrant 1 of Fig. 1) shows that, in general, those who responded to the survey in the municipality of San Andrés de Sotavento belong to indigenous communities, have a level of participation classified as high, and a low level of sanitation. A negative behavior to this, or quite the opposite, is found in the lower right

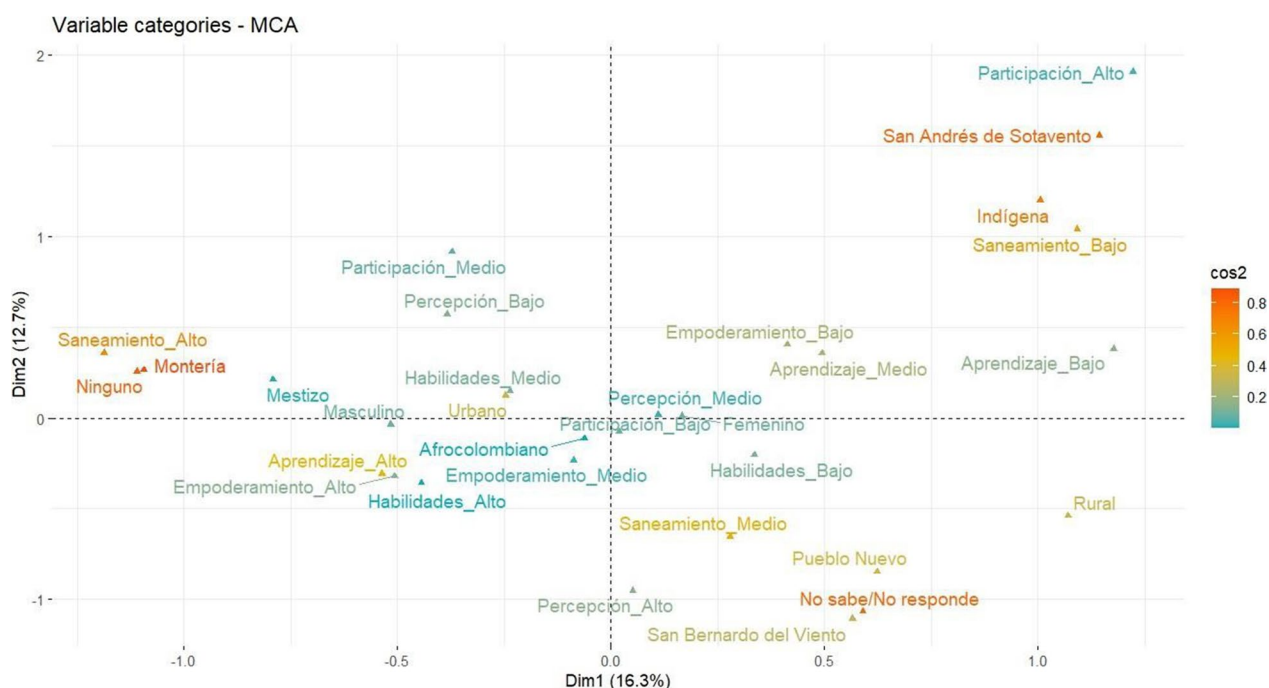


Fig. 1 Association between categories and dimensions 1 and 2

part of dimension 1 (quadrant 2), where it was identified that mainly in the municipalities of Pueblo Nuevo, San Bernardo del Viento, and San, there is a high level of perception, a medium level of sanitation, and they do not know or do not respond about their ethnic group. At the bottom of dimension 2 (quadrant 3 of Fig. 1), we see individuals with high levels of empowerment, learning, and skills, levels that are more frequently found in males and individuals of Afro-Colombian ethnicity.

In Fig. 2, it can be seen at the top of dimension 2 that a low level of empowerment is associated with low levels of learning and perception and with the mestizo ethnicity. Meanwhile, at the bottom left of dimension 3, a high level of empowerment tends to be associated with high levels of skills, perception, and learning mainly. This confirms certain behaviors found previously, where a low level of empowerment is associated with low levels of learning and skills.

Ordinal regression

An ordinal regression model was adjusted, with three levels (1: Low, 2: Medium, and 3: High) included as predictor variables: "Perception," "Participation," Knowledge and other dimensions evaluated. However, when adjusting the model, only the adjusted variables were ultimately statistically significant, while knowledge and sanitation as dimensions showed no relevant relationship and were therefore not retained in the final model. This model is represented mathematically through the cumulative logit function.

Model coefficients

The t-values were calculated for the perception ($t = 10.86$) and participation ($t = 7.17$) variables, indicating that both predictors are statistically significant at a 5% level ($p\text{-value} < 0.05$). Furthermore, the effect in both cases is positive, suggesting that as levels of perception and participation increase, so does the probability that a person will be at a higher level of empowerment.

Intercept – Cut-off points between categories

Two cut-off points were identified in the ordinal model: the Low/Medium point of 5.0177, which indicates the limit where the probability of being in "Low" is equal to that of being in a "Medium" or "High" level. The Medium/High point, 7.2093, indicates the limit where the probability of being in "Medium" is equal to that of being in "High." The Akaike Information Criterion (AIC) obtained in the model is 568.33, considered the best fit compared to other adjusted models (the 4 predictor variables, with three predictor variables) and also presented a residual deviance of 560.33, which suggests a good fit to the data, given that lower values of this metric indicate a better explanatory capacity of the model.

Variable significance test (ANOVA).

The analysis of variance confirmed that both variables, perception and participation, are statistically significant, with p-values below 0.001, indicating that both variables have a significant effect on the probability of empowerment within the adjusted model.

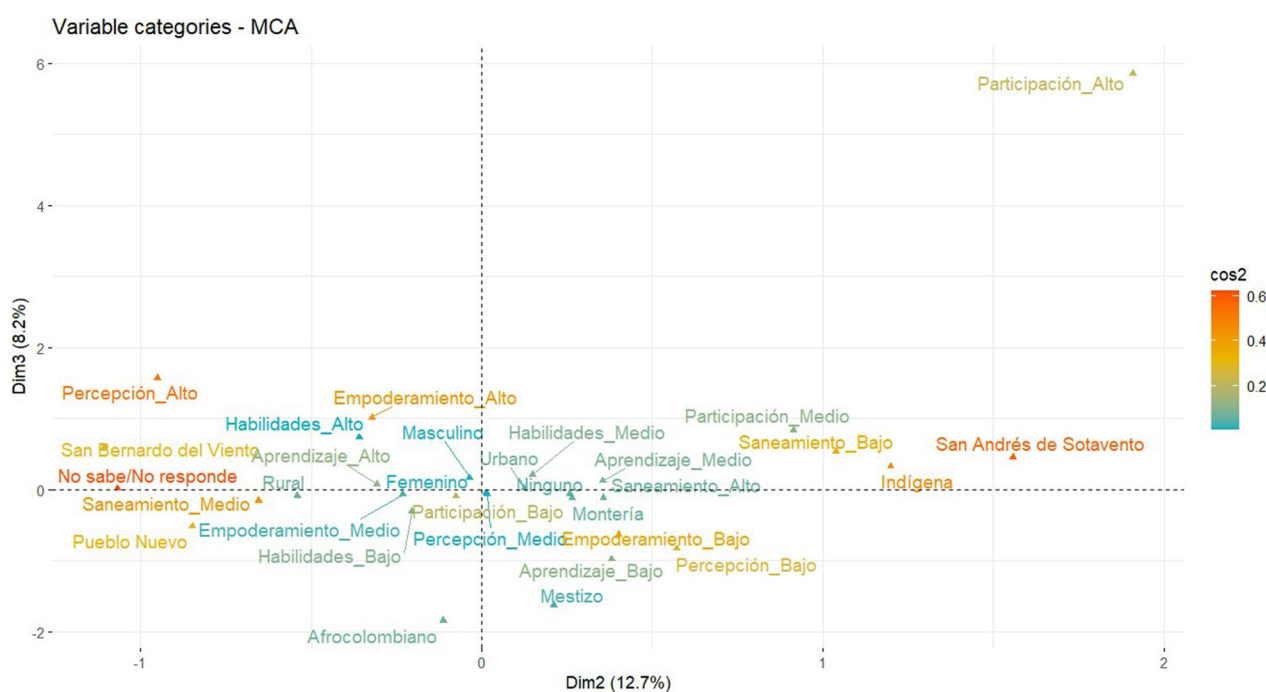


Fig. 2 Association between categories and dimensions 2 and 3

When applying the multicollinearity test (VIF – Variance Inflation Factor) to the perception and participation variables, a VIF of 1.1447 was found for both cases, given that these values were less than 5. It is concluded that there is no evidence of multicollinearity between these variables; therefore, the predictors are not highly correlated with each other and their effects can be interpreted independently.

Change in probability – Odds ratio

The adjusted model showed that the level of empowerment is significantly influenced by the variables: perception and participation. These two predictors play a central role in strengthening or weakening empowerment in the context analyzed. In concrete terms, the odds ratio (OR) associated with perception was 1.9671, indicating that when this variable increases by one unit, the probability of a person reaching a higher level of empowerment is 1.97 times greater than for those with a low level of perception. In other words, a favorable perception of capabilities and the environment in which they interact increases empowerment, which strengthens confidence and autonomous decision-making.

Similarly, participation showed a significant positive effect, with an OR of 2.0873. This indicates that as people become more actively involved in processes related to their well-being or that of their community, the probability of achieving a higher level of empowerment increases by a factor of 2.09.

Probability prediction – top 10 individuals

Table 3 shows the probability prediction of belonging to each level of empowerment (low, medium, high) for the top ten individuals in the sample, based on the adjusted ordinal regression model, thus showing how the model estimates the distribution of probabilities at different levels of the dependent variable to facilitate understanding of the models applicability in practice to individual subjects.

Table 3 Probability prediction – top 10 individuals

Level of empowerment.			
Individual	Low	Medium	High
1	0.3744	0.4683	0.1573
2	0.0324	0.1980	0.7696
3	0.3626	0.4740	0.1654
4	0.5407	0.3726	0.0867
5	0.3471	0.4792	0.1737
6	0.6984	0.2556	0.0460
7	0.2333	0.4981	0.2686
8	0.5554	0.3625	0.0821
9	0.0690	0.3298	0.6012
10	0.2333	0.4981	0.2686

Source: Own construction

In Table 3, each row of the table represents an individual, and the columns show the probabilities of belonging to each level of empowerment. For example, individual 1 has a 37.44% probability of being in the low level, a 46.83% probability of being in the medium level, and a 15.73% probability of reaching a high level. In contrast, individual 2 has a very low probability of belonging to the low level (3.24%) and a high probability of being in the high level (76.96%). Overall, the results show that the study subjects have a 39.83% probability of having low.

empowerment, a 32.59% probability of reaching a medium level of empowerment, and a 27.48% probability of being in the high level of empowerment. This means that people who enjoy better sanitation conditions in their homes, have a higher level of knowledge about the disease, and have a higher perception of participation, governance, and communication in prevention and control programs are more likely to be empowered than those with lower levels of the same characteristics or factors. In order to understand more specifically the behavior of the probabilities of the level of empowerment of the study subjects, it was found that the "Low" category has a minimum probability of 0.0010 and a maximum of 0.9750, with a median probability of 0.3744. Likewise, the "Medium" category has a minimum probability of 0.0079 and a maximum of 0.4989, with a median probability of 0.3625. With regard to the "High" category, a minimum probability of 0.0029 was obtained, reaching a maximum of 0.9911 and a median probability of 0.1573.

As can be seen, in the "Low" category, 50% of the study subjects have probabilities lower than 0.3794 of belonging to this category. Similarly, in the "Medium" category, 50% of respondents have probabilities greater than 0.3625, and in the "High" category, 50% have probabilities lower than 0.1573, which indicates a low representation of this category within the sample analyzed.

As a conclusion of the model applied, it was identified that it is statistically significant, given that both perception and participation affect the probability of empowerment. In particular, it was found that greater perception increases the probability of being at a high level of empowerment (OR=1.97; p-value<0.0001), while when a person has a higher level of participation, the probability of being in a higher category of empowerment also increases (OR=2.09; p-value<0.0001). No multicollinearity problems were found, as VIFs were less than 5, and the predicted probabilities suggest that most individuals are in the "Medium" category.

Finally, the Brant test was applied to verify compliance with the assumption of proportional odds in the ordinal regression model, which supports the validity of the methodological approach used.

Table 4 Applications of the ordinal model

Case	Scores		Probability		
	Perception	Participation	Low	Medium	High
1	1	2	94.63%	4.74	0.06%
2	11	8	0.02	0.19	99.78%
3	13	0	2.24%	14.76%	83.00%
4	3	9	2.57%	16.53%	80.90%

Source: Own construction

Brant test for the ordinal model

assesses the proportionality of odds, yielded favorable results for the model. The overall goodness-of-fit test (Omnibus) reported a p-value of 0.11, indicating that there is no evidence to reject the null hypothesis, i.e., the odds assumption in the model is satisfied. Likewise, the analysis of the individual variables showed that both perception (p-value=0.05) and participation (p-value=0.18) do not violate the assumption of proportional odds. These results confirm that the ordinal regression model is adequate and valid for interpretation.

For this case, the adjusted model is:

$$\log(P(Y \leq 1)/P(Y > 1)) = 5.0177 + 0.676512 \cdot \text{Perception} + 0.7359 \cdot \text{Participation}$$

$$\log(P(Y \leq 2)/P(Y > 2)) = 7.2093 + 0.6765 \cdot \text{Perception} + 0.7359 \cdot \text{Participation}$$

Where:

$P(Y \leq 1)$ is the probability that empowerment will be "Low".

$P(Y \leq 2)$ is the probability that empowerment will be "Medium" or lower. $P(Y > 1)$ is the probability that empowerment will be "Medium" or higher.

$P(Y > 2)$ is the probability that empowerment will be "High".

$\beta_1 = 0.6765$: A higher level of sanitation increases the probability of being at a high level of empowerment.

$\beta_2 = 0.7359$: A higher level of knowledge increases the probability of being at a high level of empowerment.

5.0177 and 7.2093 are the cut-off thresholds (intercepts) that separate the ordinal categories.

Applications of the ordinal model

Table 4 illustrates how changing the levels of these predictor variables changes the estimated probabilities of an individual being at different levels of empowerment.

Table 4 presents scenarios for applying the adjusted ordinal model, where each row of the result represents a specific scenario with given values of the independent variables.

In case 1, it is more likely that the individual has a low level of empowerment, considering that perception is assigned a score of 1, which is classified as low, and participation is assigned a score of 2, which is also classified as low. In case 2, with a perception score of 11, which is

classified as high, and a participation score of 8, which is also classified as high, there is an increase in the probability of having medium empowerment with a percentage of 99.78%. This reinforces the result that greater empowerment is found with greater perception and participation.

For the third case, there is a higher probability of medium empowerment with a percentage of 14.76%, obtaining a score of 13 classified as high in perception and a score of 0 classified as low in participation, in turn decreasing the percentage of having high empowerment to a percentage of 83.00%. Finally, if we take a score of 3 classified as low perception and a score of 9 classified as high participation, we obtain a result with a higher probability of high empowerment with a percentage of 80.90%.

In general, these results allow us to establish three empowerment groups. First, a population group residing mainly in the urban area of the municipality of Montería, the departmental capital, with a predominance of males, a high level of sanitation, a high level of learning, a medium level of dengue prevention and control skills, and a high level of empowerment and perception.

Another group of individuals, residing in the municipalities of Pueblo Nuevo and San Bernardo del Viento, with medium and low levels of skills and participation, have a high level of perception and a medium level of sanitation. and a third group, made up of individuals who identify themselves in their responses as belonging to the Zenú indigenous ethnic group, located in the municipality of San Andrés de Sotavento, mostly residing in rural areas, have a high level of participation, low sanitation, and a medium level of empowerment.

Discussion

This research set out to examine how participation and governance are related as possible factors in the empowerment of the population affected by dengue in the department of Córdoba, Colombia. It was found that both perceptions of communication, education, and governance processes and participation influence the likelihood of empowerment. In general, it was found that greater perception increases the probability of being at a high level of empowerment, and that when a person has a higher level of participation, the probability of being in a higher category of empowerment also increases.

The dimensions of family empowerment (participation, communication, and governance) have been explored from different perspectives in relation to dengue. For example, the study by Ririn and Pramono [21] concludes that the higher the level of psychological empowerment and sense of community, the greater the likelihood of participating in dengue care programs [22]. Furthermore, supplementing knowledge about dengue increases motivation to participate in control and prevention programs

[25], and as community environmental management strategies are implemented, greater progress is made in terms of interventions [23].

Other variables and/or auxiliary categories considered in this study were geographical aspects (municipality and urban and rural areas), ethnicity (indigenous and Afro-descendants), level of environmental sanitation, and level of learning about the disease. Although these factors influence each other and the different levels of the established empowerment groups, the differences that separate them are not strictly geographical in nature, nor are they ethnic and cultural characteristics, but rather the sanitary conditions recorded in their territories.

In line with the above, in Colombia, sanitary conditions are recognized as one of the indicators for social stratification, with the aim of applying differential rates for residential services, subsidies, and other government assistance. This characteristic could explain the results found with respect to the odds ratio (OR) of participation, perception, and empowerment, since access to and quality of such services as drinking water supply and wastewater management are concentrated in urban centers, mainly benefiting the highest social strata compared to those classified at the lowest level, generally located in urban peripheries.

Similarly, variability in sanitary conditions also has a direct impact on the reasons why individuals or communities participate in collective affairs and on the favorable perception of their own capabilities and the environment in which they interact, increasing their confidence and autonomy in decision-making. In this sense, poor sanitation generates health problems and limits development, affecting their capacity for action and participation for empowerment. Similar findings were reported by Asnita [24] and Putri and Ardinigrum [25].

Therefore, it is understandable to find in our study that stratum 1 is 3.3 times more likely to have medium empowerment, compared to stratum 2, which is 3.6 times more likely to have low empowerment. The same is true of the relationship between knowledge of the disease and levels of empowerment, where socioeconomic stratum 1 ($p = 0.0050$) is 2.89 times more likely to have a medium level of knowledge, in contrast to stratum 2, which is 3.75 times more likely to have a low level of knowledge.

Regarding the incidence of geographical and ethnic factors and the level of learning on community empowerment, even though these did not specify the differentiation of the groups established in this study, they do complement the explanation of the predictive factors of empowerment. Some studies support this assessment, concluding that the implementation of strategies to reduce the high seroprevalence rates of dengue must consider the location of the home and ethnicity as

determining factors [26], as well as knowledge about the disease [27].

However, the incorporation of these variables into dengue care processes should cease to be a technical matter because, according to Valencia [18], the particularities, knowledge, and cultural practices of the different groups in Córdoba point to opportunities for empowerment to the extent that multivariate and dialogical scenarios are produced that take into account family experiences, face-to-face interaction, and the recognition of natural forms of participation and governance. This has made it possible to tackle dengue through community ties, the exchange of medicinal plants, solidarity, and community mobilization, among other means.

From the point of view of perceptions of institutional communication and governance of dengue prevention and control programs and their impact on empowerment, there are certain differences with the findings reported by Benítez et al., [28] and Marrufo [29], who identified a predominant perception of state responsibility (62% and 69%, respectively), and with Torres et al., [30], whose participants in their responses hold these administrative entities in the health sector in the territories (33%) and the state (30%) responsible, while in our case, this responsibility is mainly attributed to the family, despite the recognition of the lack of adequate institutional supervision and monitoring of the aforementioned prevention and control programs.

In this regard, it is valid to consider an alternative form of communication as proposed by the Integrated Management Strategy for Dengue Prevention and Control (EGI-Dengue), in terms of taking advantage of social media and technology in the management of this strategy, which requires specific studies with different approaches aimed at its implementation, as suggested by Martínez et al. [31].

In line with the above, there is a need to reflect on a change in the concept and institutional function assigned to communication in the management processes of these programs, which go beyond the mere informative sphere and transcend the human potential of the actors involved, in which communities fully exercise their rights by contributing and learning from their own experience and capacities, according to Shafique et al., [32]. For these authors, the effectiveness of "insider" interventions depends on the creation of spaces for community dialogue, the identification of local needs, and the participatory development of information, education, and communication (IEC) materials. Vargas et al. [33] share this opinion, arguing that the active involvement of the community favors the eradication of vector breeding sites and enhances the sustainability of prevention strategies.

These considerations reinforce the concept of Health for All (HFA), promoted by the World Health

Organization (WHO), which conceives physical, mental, and social well-being as a goal for the whole of society, requiring coordinated action between governments, businesses, organizations, and citizens [34]. This assessment in turn reaffirms the hypothesis that those who have greater opportunities for participation are more likely to have a better perception of the communication and governance of dengue prevention and control programs and, consequently, tend to be more involved in empowerment actions.

Among the limitations of the study, which may limit the scope of its interpretation and validity, is the absence of other types of questions and inquiry techniques involving organizational aspects of participation and various forms of social representation, such as ancestral knowledge, beliefs, practices, and customs that give meaning to dengue prevention and control activities in this region of the country, which invite further exploration of the topic from comprehensive and combined perspectives, especially in endemic and absolute poverty contexts such as the department of Córdoba, Colombia.

Nevertheless, we consider the findings of this study to be valuable and highly relevant, because they provide information that can be useful for public policy makers with a differential and territorial approach, where the concept of participation in health transcends the simple involvement of actors to successfully develop health programs [21, 35], mainly within the framework of the administrative decentralization of the health sector, through which powers and functions are delegated to territorial entities with relative autonomy for the execution of such programs, as stipulated in Law 715 of 2001, Law 1751 of 2015, and Decree 0858 of 2025 issued by the Ministry of Health and Social Protection of Colombia, which adopts the Preventive, Predictive, and Resolutive Health Model as a national policy to transform the health system in the country [36–38].

Conclusion

In accordance with the proposed objective, it was found that both perception and participation influence the probability of empowerment, establishing that greater perception increases the probability of being at a high level of empowerment (OR = 1.97; p -value < 0.0001), while when a person has a higher level of participation, the probability of being in a higher category of empowerment also increases (OR = 2.09; p -value < 0.0001).

Overall, it was determined that the study subjects have a 39.83% probability of having low empowerment, a 32.59% probability of reaching a medium level of empowerment, and a 27.48% probability of being at a high level of empowerment.

Based on the results obtained, dengue should be considered not only as a communicable disease or

epidemiological phenomenon, but also as a social and political problem that requires participatory, intersectoral, and territorialized models, recognizing the importance of participation and governance as potential factors in the development of human capacities and community empowerment, promoting horizontal communication processes that articulate ancestral knowledge and scientific knowledge according to the particularities of the territories.

Consequently, it is in this regulatory context, mainly in light of Resolution 1789 of 2025 of the Ministry of Health and Social Protection, which establishes the territorialization of the health system in Colombia, dividing the country into 10 regions and 119 functional subregions for the comprehensive territorial management of public health, where networks for primary care, control, and prevention of dengue could begin to be built, starting with the identification not only of risk factors and treatment of the disease, but also of potential factors for the development and empowerment of families and communities in terms of education, planning, research, communication, and governance, with the support and coordination of other sectors and stakeholders, such as universities, schools, hospitals, health centers, institutions, and local and departmental territorial entities.

Author contributions

All authors contributed to the conception or design of the work. CV substantially reviewed the manuscript. JO performed the data analysis. All authors reviewed and approved the manuscript.

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Data availability

Requests for data/analysis code can be made via the corresponding author.

Declarations

Competing interests

The authors declare no competing interests.

Research involving human statements

The study adhered to the ethical guidelines for research involving human subjects stipulated in Resolution 8430 of 1993, and was declared a minimal-risk study.

Informed consent

Written informed consent was obtained from all participants, and the study was submitted to the review committee at the University of Córdoba, receiving approval under Code FE-11-19. ethical criteria ethical criteria ethical criteria.

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